

Skill Builder Pack 2 - Cleaning Data in Excel Using Power Query

Purpose

After data has been imported into Excel, it is rarely ready to use immediately.

Data exported from systems often contains issues that can affect analysis. These may include:

- Incorrect formatting
- Blank rows or columns
- Extra spaces in text
- Inconsistent column names
- Incorrect data types
- Duplicate records

If these issues are not addressed, they can lead to incorrect analysis or misleading results.

For example, a column containing dates stored as text cannot be used correctly in time-based analysis, and names containing extra spaces may appear as separate entries even though they refer to the same person.

For this reason, data analysts spend a significant amount of time cleaning and preparing datasets before analysis begins.

Excel provides a powerful tool called Power Query that allows datasets to be cleaned in a structured and repeatable way.

This skill builder focuses on learning how to prepare datasets so that they are consistent, accurate and suitable for analysis.

Learning Outcomes

By the end of this skill builder you should be able to:

- Open imported datasets in the Power Query Editor
- Identify common data quality issues
- Correct data types
- Rename columns clearly
- Remove blank or unnecessary rows
- Clean text fields using trimming tools
- Ensure datasets use consistent identifiers

These skills are part of the data preparation stage of the analysis process.

In case you need it

Clean It Up - TRIM & CLEAN

Chorus

Trim it down, clean it out
Make the messy data count
Split it up, set it right
Now the picture's clear in sight

Verse 1

Spaces hiding at the ends
Break the way the data blends
Use TRIM to cut them clean
Now the names are what they mean

Verse 2

Hidden marks you cannot see
Use CLEAN to set data free
No more errors, no disguise
Now it works when you analyse

Bridge

Step by step, it's all recorded
Every fix is stored and sorted
Refresh the data, watch it flow
Same clean path, it runs on its own

Make It Match - TEXT

Chorus

Upper, lower, make it match
Stop the data breaking batch
Same words, same every time
Now it flows and aligns

Verse 1

Ali, ali, ALI too
Looks the same but not to you
UPPER brings it all in line
Lower works just as fine

Verse 2

Consistency is what you need
Or your joins will never read

Same format, same design
That's how datasets combine

Bridge

Different forms but same intent
Now they match, the noise is spent
One clean standard, strong and clear
Now your joins will work right here

Break It Apart - SPLITTING CELLS

Chorus

Split it out, break it down
Turn one column into two now
Comma, space, or slash between
Now the structure's sharp and clean

Verse 1

Full names stuck in one long field
Hard to use, no proper yield
Split by delimiter, choose your sign
Now first and last are clearly defined

Verse 2

Emails, codes, or address lines
All can split along the signs
Once it's done, you'll clearly see
Better data quality

Bridge

One field holding far too much
Now refined with just one touch
Break it down so it can grow
That's the way clean data flows

Strip It Back - BLANKS AND DUPLICATES

Chorus

Cut the blanks, clear the noise
Leave the data you can trust and use
No more gaps, no empty rows
Now your analysis can flow

Verse 1

Rows with nothing, taking space
Dragging down the dataset

Remove blanks, keep what's real
Now your numbers truly feel

Verse 2

Duplicates that double count
Throw your totals way off amount
Remove duplicates, make it true
One clean record coming through

Bridge

Less is more when data's clean
Only keep what should be seen
Strip it back and make it right
Now your outputs hold up tight

One Version of Truth

Chorus

One version, one clear line
No repeats, no second time
Clean it once and make it true
So the data works for you

Verse 1

Duplicates can trick the eye
Make your totals multiply
Strip them out and make it right
Now your figures hold up tight

Verse 2

Clean and structured, ready now
Every step you've shown it how
Run it once, it stays that way
Ready every single day

Bridge

Trust the process, trust the flow
Every step you now can show
Clean foundation, strong and true
Everything depends on you

Starting Point

You should already have an Excel workbook containing the datasets imported during Skill Builder Pack 1.

These datasets include:

- Attendance records
- Student details
- Engagement scores
- Withdrawal records

Each dataset currently exists as a separate table in Excel.

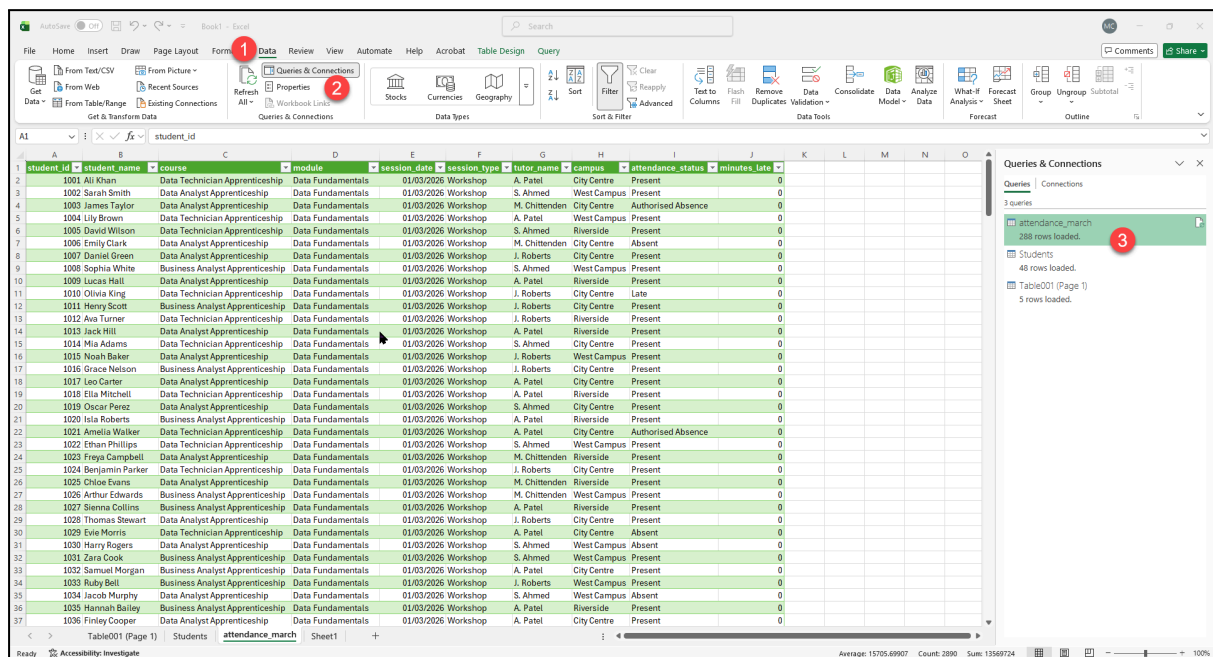
In this skill builder you will open each dataset in Power Query and prepare it for later analysis.

Opening Power Query

To clean a dataset, you must first open it in Power Query.

Follow these steps:

1. Select the Data tab in Excel
2. Open **Queries & Connections**
3. Locate the dataset you want to edit



4. Double-click the query name

This opens the Power Query Editor.

The editor allows you to perform data cleaning tasks such as:

- Changing formats
- Removing rows
- Editing column names
- Transforming text

Every change you make is recorded as a step. This means that the same cleaning process can be automatically repeated whenever the data is refreshed.

Skill 1 – Understanding Data Types

Each column in a dataset has a data type, which tells Excel how the values should be interpreted.

Common data types include:

Data Type	Example
Text	Names or categories
Whole Number	Student ID numbers
Decimal Number	Scores or measurements
Date	Session dates

If the data type is incorrect, Excel may not analyse the data properly.

For example:

- Dates stored as text cannot be used in date calculations.
- Numbers stored as text cannot be summed or averaged.

Practice Activity

Open the attendance dataset in Power Query.

Check the following columns by clicking on the icon to the left of the column name:

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: student_id, student_name, course, module, session_date, session_type, tutor_name, campus, and attendance_status. The 'student_id' column is highlighted, and its data type is set to 'Whole Number'. The 'session_date' column is highlighted, and its data type is set to 'Date'. The 'attendance_status' column is highlighted, and its data type is set to 'Text'. The 'Properties' pane on the right shows the 'Changed Type' step applied to the 'attendance_status' column.

- student_id
- session_date
- attendance_status

Ensure that:

- student_id is set to Whole Number
- session_date is set to Date
- attendance_status is set to Text

If a column has the wrong type, change it using the data type icon next to the column name.

Skill 2 – Renaming Columns

Column names should be clear and easy to understand.

When datasets come from different systems, column names may be inconsistent or unclear.

For example:

Poor Name	Improved Name
ID	student_id
Date	session_date
Status	attendance_status

Clear naming helps ensure datasets can be understood and combined correctly.

Practice Activity

Review column names in each dataset.

Make sure that:

- Column names are clear
- Similar fields use consistent naming
- Important identifiers (such as student_id) have identical names across datasets

Skill 3 – Removing Blank Rows

Exported datasets often contain blank rows that do not contain any data.

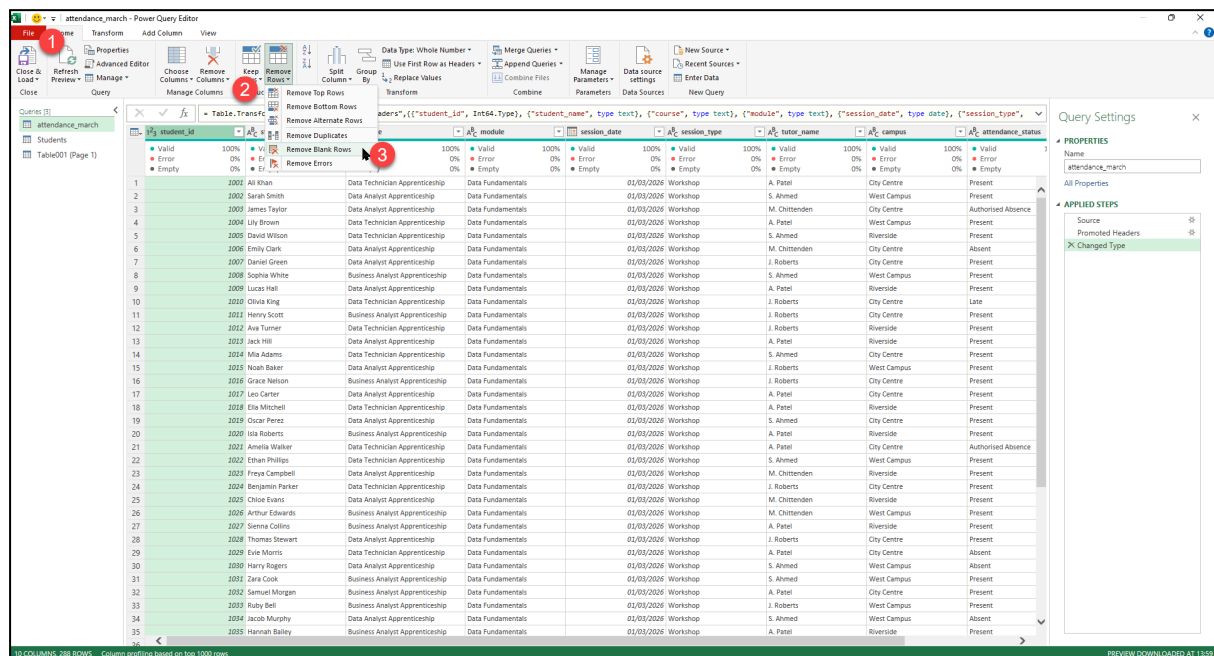
These rows can cause problems during analysis and may affect calculations or charts.

Power Query makes it easy to remove them.

Practice Activity

Open the attendance dataset.

1. Select the Home tab in Power Query
2. Choose Remove Rows
3. Select Remove Blank Rows



Check that only rows containing real records remain.

Skill 4 – Cleaning Text Fields

Text data often contains unwanted spaces.

For example, the following values may appear identical but are treated differently by Excel:

Ali Khan
Ali Khan
Ali Khan

These variations can cause duplicate entries or incorrect grouping.

Power Query provides tools to clean text values.

Practice Activity

Open the **Students** dataset.

Select the following columns:

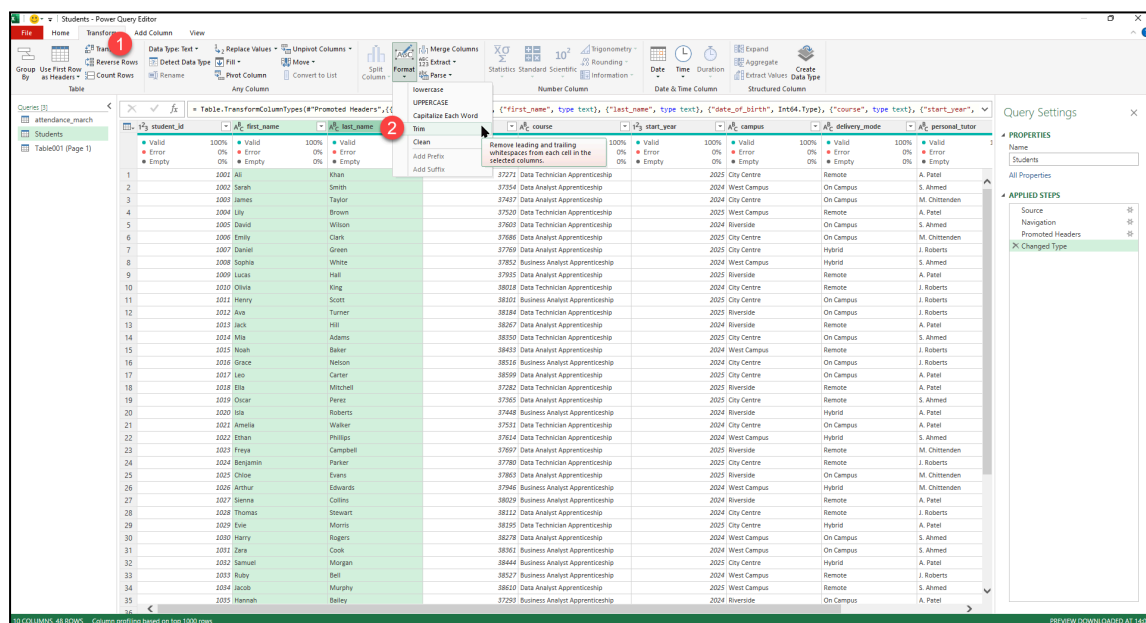
- first_name
- Last_name

If you want, press **SHIFT** to multi-select.

Then:

1. Go to the Transform tab
2. Choose Format → Trim

This removes spaces at the beginning and end of text values.



Skill 5 – Checking Data Consistency

Datasets often need to be combined later using a shared identifier.

In this case, the key identifier is:

student_id

If this field is inconsistent across datasets, the data cannot be joined correctly.

Practice Activity

Review the student_id column in all datasets.

Check that:

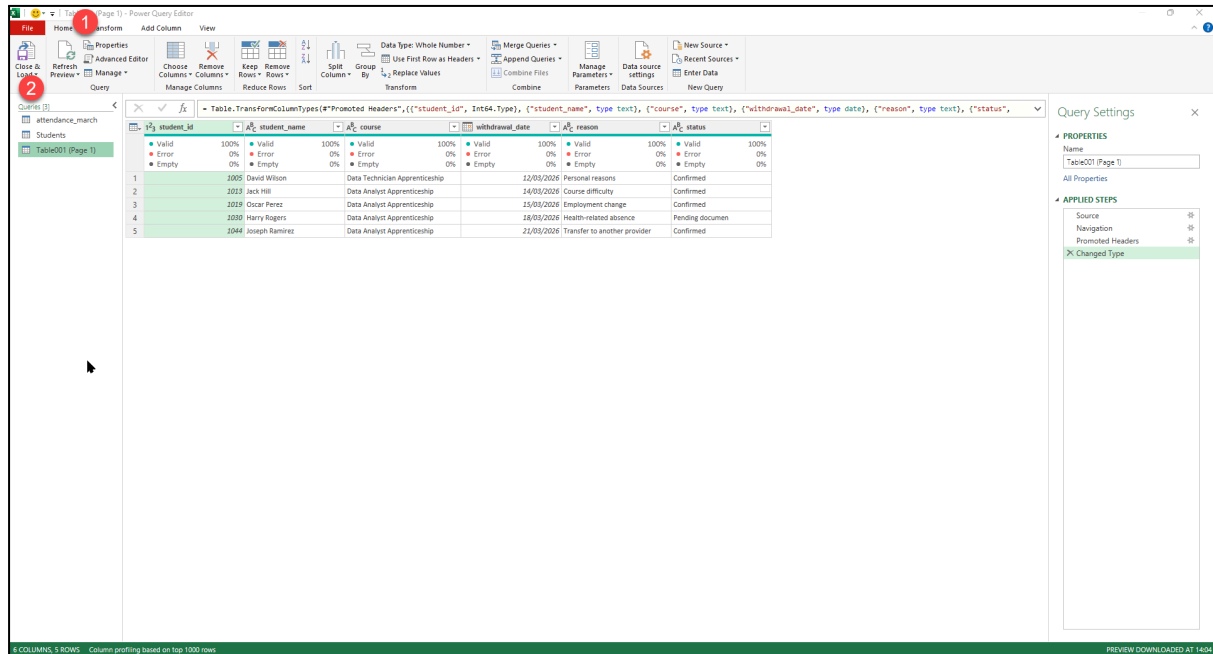
- The column name is identical
- The data type is the same
- Values appear consistent

This step ensures the datasets can later be combined successfully.

Loading Cleaned Data Back into Excel

Once you have completed the cleaning steps:

1. Select Home
2. Click Close & Load



The cleaned datasets will be loaded back into Excel.

All transformation steps are saved.

This means that if the source data is updated later, Excel can simply refresh the queries and repeat the cleaning automatically.

Key Concept

Cleaning data is a critical stage of data analysis.

Analysts rarely begin with perfect data. Instead, the typical workflow is:

Import → Clean → Combine → Analyse → Visualise

In this skill builder you have focused on the Clean stage.

Preparing datasets carefully helps ensure that any later analysis is accurate and reliable.